Ultraviolet Sobolev Ill-Posedness for a Fourth-Order Wave Equation

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Abstract

We prove that the constant-coefficient fourth-order wave equation $(\Box - \Box^2 - \Delta^2)u = 0$, with $\Box = -\partial_t^2 + \Delta$, has no bounded positive-time evolution operator on any finite-order Sobolev four-jet space, even with arbitrary fixed finite loss of spatial derivatives. The proof derives the exact temporal roots from the dispersion relation $D(z, q) = (z - q) - (z - q)^2 - q^2$, establishes high-frequency growth $\operatorname{Re} s(k) = \sqrt{(\sqrt{2} - 1)/2}\,|k| + O(|k|^{-1})$, and constructs exact solution sequences on both a flat torus and the whole space. We also state the extension to the strict positive-curvature coefficient regime of the homogeneous S004 linearization. These are linear, constant-background conclusions, not nonlinear norm-inflation or finite-lifetime theorems.

1 Equation and precise statement

Let $d \geq 1$, and let $M = \mathbb{T}^d$ or $M = \mathbb{R}^d$, where $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$. Define

$$\Box = -\partial_t^2 + \Delta, \qquad (\Box - \Box^2 - \Delta^2)u = 0. \tag{1}$$

Expanding gives the real-coefficient equation

$$\partial_t^4 u + (1 - 2\Delta)\partial_t^2 u - \Delta u + 2\Delta^2 u = 0. \tag{2}$$

For $r \in \mathbb{R}$, introduce the four-jet Sobolev space

$$X_r(M) = H^{r+3}(M) \times H^{r+2}(M) \times H^{r+1}(M) \times H^r(M), \qquad \|U\|_{X_r}^2 = \sum_{j=0}^{3} \left\| \partial_t^j u \right\|_{H^{r+3-j}}^2, \tag{3}$$

where $U(t) = (u, \partial_t u, \partial_t^2 u, \partial_t^3 u)(t)$.

Theorem 1 (Unbounded positive-time Sobolev evolution)**.** *For every $r \in \mathbb{R}$, every fixed finite $\ell \geq 0$, and every $t_0 > 0$, classical evolution for (2), defined on the smooth Fourier-mode or Fourier-packet data constructed below, admits no bounded linear extension*

$$S(t_0) : X_{r+\ell}(M) \longrightarrow X_r(M).$$

Indeed, there exist smooth exact solutions u_n such that

$$\|U_n(0)\|_{X_{r+\ell}} \longrightarrow 0, \qquad \|U_n(t_0)\|_{X_r} = 1.$$

In particular, standard Sobolev continuity fails even if a fixed finite number of spatial derivatives is lost.

2 Exact characteristic roots

For $u = e^{i(k\cdot x-\omega t)}$, set $z = \omega^2$ and $q = |k|^2$. Then $\Box u = (z-q)u$ and $\Delta^2 u = q^2 u$, so (1) yields

$$D(z,q) = (z-q) - (z-q)^2 - q^2 = 0. \tag{4}$$

Expanding and changing the overall sign,

$$z^2 - (1+2q)z + q + 2q^2 = 0.$$

Its discriminant is exactly

$$(1+2q)^2 - 4(q+2q^2) = 1 - 4q^2.$$

Therefore

$$z_\pm(q) = q + \frac{1 \pm \sqrt{1-4q^2}}{2}. \tag{5}$$

For $q > \frac{1}{2}$, the roots are nonreal conjugates:

$$z_\pm(q) = q + \frac{1}{2} \pm \frac{i}{2}\sqrt{4q^2-1}. \tag{6}$$

3 Growing temporal roots and asymptotics

Write $z_+(q) = A(q) + iB(q)$, where

$$A(q) = q + \frac{1}{2}, \qquad B(q) = \frac{1}{2}\sqrt{4q^2-1} > 0.$$

Choose its square root $\omega_+(q) = \alpha(q) + i\sigma(q)$, with $\alpha, \sigma > 0$. By taking real and imaginary parts of $\omega_+^2 = z_+$,

$$\alpha^2 - \sigma^2 = A, \qquad 2\alpha\sigma = B, \qquad \alpha^2 + \sigma^2 = \sqrt{A^2+B^2}.$$

Thus

$$\sigma(q) = \sqrt{\frac{\sqrt{A(q)^2+B(q)^2} - A(q)}{2}} = \sqrt{\frac{\sqrt{2q^2+q} - q - \frac{1}{2}}{2}}. \tag{7}$$

The exponent $s(q) = -i\omega_+(q) = \sigma(q) - i\alpha(q)$ satisfies $\operatorname{Re} s(q) = \sigma(q) > 0$, since $e^{-i\omega_+ t} = e^{s(q)t}$.

As $q \to \infty$,

$$\sqrt{2q^2+q} = \sqrt{2}\,q + \frac{\sqrt{2}}{4} + O(q^{-1}),$$

whence

$$\sigma(q)^2 = \frac{\sqrt{2}-1}{2}\,q + \left(\frac{\sqrt{2}}{8} - \frac{1}{4}\right) + O(q^{-1}), \tag{8}$$

$$\sigma(q) = \sqrt{\frac{\sqrt{2}-1}{2}}\,\sqrt{q} + O(q^{-1/2}). \tag{9}$$

Since $q = |k|^2$,

$$\boxed{\operatorname{Re} s(k) = c_{\mathrm{UV}}|k| + O(|k|^{-1}), \qquad c_{\mathrm{UV}} = \sqrt{\frac{\sqrt{2}-1}{2}} = 0.455089860562227\ldots > 0.} \tag{10}$$

Consequently, for any $0 < c_* < c_{\mathrm{UV}}$ there exists $R > 0$ such that $\operatorname{Re} s(k) \geq c_*|k|$ whenever $|k| \geq R$. Moreover $|s(k)| \asymp |k|$ at high frequency.

4 Proof on the flat torus

Proof of Theorem 1 for $M = \mathbb{T}^d$. Choose nonzero $k_n \in \mathbb{Z}^d$ with $|k_n| \to \infty$, and take the growing root $s_n = s(k_n) = \sigma_n + i\nu_n$, where $\sigma_n > 0$. For each real $a_n > 0$, the function

$$u_n(t,x) = a_n \operatorname{Re}(e^{s_n t + ik_n \cdot x}) = a_n e^{\sigma_n t} \cos(k_n \cdot x + \nu_n t) \tag{11}$$

is an exact, real-valued, smooth solution of (2). Differentiation gives

$$\partial_t^j u_n = a_n \operatorname{Re}(s_n^j e^{s_n t + ik_n \cdot x}), \qquad j = 0, 1, 2, 3.$$

Let $C_{\mathbb{T}} = \|\cos(k_n \cdot x)\|^2_{L^2(\mathbb{T}^d)} > 0$, which is independent of the nonzero integer vector k_n. For any real p, phase invariance of a single Fourier mode yields

$$\left\| \operatorname{Re}(s_n^j e^{ik_n \cdot x}) \right\|^2_{H^p} = C_{\mathbb{T}} |s_n|^{2j} (1 + |k_n|^2)^p.$$

Set

$$N_{n,r}^2 = C_{\mathbb{T}} \sum_{j=0}^{3} |s_n|^{2j} (1 + |k_n|^2)^{r+3-j} > 0. \tag{12}$$

Then, exactly,

$$\|U_n(t)\|_{X_r} = a_n e^{\sigma_n t} N_{n,r}. \tag{13}$$

Fix $t_0 > 0$ and choose

$$a_n = \frac{e^{-\sigma_n t_0}}{N_{n,r}}.$$

It follows that

$$\|U_n(t_0)\|_{X_r} = 1. \tag{14}$$

For any $\ell \geq 0$, each Sobolev component in (12) gains the same factor $(1 + |k_n|^2)^\ell$ at the level of squared norms. Therefore

$$N_{n,r+\ell} = (1 + |k_n|^2)^{\ell/2} N_{n,r},$$

and hence

$$\|U_n(0)\|_{X_{r+\ell}} = e^{-\sigma_n t_0} (1 + |k_n|^2)^{\ell/2} \longrightarrow 0, \tag{15}$$

because $\sigma_n \geq c_* |k_n|$ for all sufficiently large n, and an exponential dominates every fixed polynomial.

If a bounded extension $S(t_0) : X_{r+\ell} \to X_r$ existed, (14)–(15) would imply

$$1 = \|S(t_0) U_n(0)\|_{X_r} \leq \|S(t_0)\| \, \|U_n(0)\|_{X_{r+\ell}} \longrightarrow 0,$$

a contradiction. □

5 Proof on the whole space

Proof of Theorem 1 for $M = \mathbb{R}^d$. The plane waves above are not in $L^2(\mathbb{R}^d)$, so replace them by Fourier packets. Let e_1 be the first coordinate vector, set $k_n = ne_1$, and choose a nonzero real-valued $\phi \in C_c^\infty(B(0,1))$. For large n, the growing characteristic root $s(\xi)$ is smooth in $B(k_n, 1)$, and

$$\operatorname{Re} s(\xi) \geq c_* |\xi| \qquad (\xi \in B(k_n, 1)).$$

Define the exact complex solution v_n by

$$\widehat{v}_n(t,\xi) = \phi(\xi - k_n)e^{s(\xi)t},$$

and take its real part $u_n = \operatorname{Re} v_n$. The reflected Fourier supports contributing to u_n are disjoint for large n. Consequently, Plancherel's theorem and the uniform pointwise bound on $\operatorname{Re} s$ give

$$\|U_n(t_0)\|_{X_r} \geq e^{c_*(n-1)t_0} \|U_n(0)\|_{X_r}. \tag{16}$$

On the same frequency supports, $|\xi| \leq n+1$. Thus

$$\|U_n(0)\|_{X_{r+\ell}} \leq (1+(n+1)^2)^{\ell/2} \|U_n(0)\|_{X_r}. \tag{17}$$

Combining (16) and (17),

$$\frac{\|U_n(t_0)\|_{X_r}}{\|U_n(0)\|_{X_{r+\ell}}} \geq \frac{e^{c_*(n-1)t_0}}{(1+(n+1)^2)^{\ell/2}} \longrightarrow \infty.$$

Rescale the solutions so that the numerator is exactly 1. Their initial $X_{r+\ell}$-norms then tend to zero, contradicting any bounded extension from $X_{r+\ell}(\mathbb{R}^d)$ to $X_r(\mathbb{R}^d)$. □

6 General homogeneous-sector coefficient extension

The same mechanism extends to the dispersion polynomial

$$\mathcal{D}(z,q) = (z-q) - m^2 - K(z-q)^2 + Gq(z-q) - Tq^2, \tag{18}$$

under the conditions

$$K > 0, \qquad T > 0, \qquad B := 4KT - G^2 > 0, \qquad m^2 \in \mathbb{R} \text{ fixed}. \tag{19}$$

Proposition 2 (Strict-coefficient ultraviolet growth)**.** *For (18)–(19), the growing temporal root obeys*

$$\operatorname{Re} s(k) = c_{\mathrm{UV}}(K,G,T)|k| + O(|k|^{-1}), \tag{20}$$

where

$$a = \frac{2K+G}{2K}, \qquad b = \frac{\sqrt{4KT-G^2}}{2K} > 0, \qquad c_{\mathrm{UV}}(K,G,T) = \sqrt{\frac{\sqrt{a^2+b^2}-a}{2}} > 0. \tag{21}$$

Accordingly, the conclusion of Theorem 1 applies to each scalar constant-coefficient sector governed by (18).

Proof. With $y = z - q$, solving the quadratic equation in y yields

$$z_\pm(q) = q + \frac{1+Gq \pm \sqrt{1-4Km^2+2Gq-Bq^2}}{2K}.$$

Since $B > 0$, as $q \to \infty$ the square root is

$$\sqrt{1-4Km^2+2Gq-Bq^2} = i\sqrt{B}\, q + O(1)$$

on the branch with positive imaginary part. It follows that

$$z_\pm(q) = q(a \pm ib) + O(1).$$

The square root of $a+ib$, with $b > 0$, has positive imaginary part $\sqrt{(\sqrt{a^2+b^2}-a)/2}$. Taking $\omega^2 = z_+$ and $s = -i\omega$ therefore proves (20). Positivity is strict because $b > 0$ implies $\sqrt{a^2+b^2} > |a| \geq a$. The growing eigenmodes have $|s(k)| \asymp |k|$; the torus and packet arguments above then apply verbatim. □

7 Scope of the conclusion

Remark 3 (Linearization and nonlinear evolution)**.** *The theorem is a linear, constant-coefficient result. In the corrected S004 analysis,* (18) *describes scalar blocks obtained by linearizing about a nonzero static homogeneous on-shell background with the stated coefficient conditions. Such a background on* $\mathbb{R}^d$ *need not have finite total energy. The unbounded linear maps also preclude a nonlinear flow having a bounded Fréchet derivative at that background equal to this linearization. No independent nonlinear norm-inflation theorem follows from the present argument.*

Remark 4 (Exceptional or modified regimes)**.** *The strict condition* $4KT - G^2 > 0$ *is essential for Proposition 2; degenerate symbols, zero backgrounds, inhomogeneous backgrounds, modified principal terms, and constrained completions require separate analyses. Band-limited or analytic-data evolution may have different hypotheses. The theorem assigns no numerical decay time or physical confinement lifetime.*

Source and reproducibility

This exposition expands the high-frequency argument in Theorem H4 of the foundational audit:

> R. J. Reyes, *S004 physical-health audit: constraints, ghost and ultraviolet obstruction* (2026), Foundational audit and proof source.

A symbolic/numerical verification record is available at Verification record. Those checks are supplementary: the obstruction above follows directly from the analytic Fourier solutions.